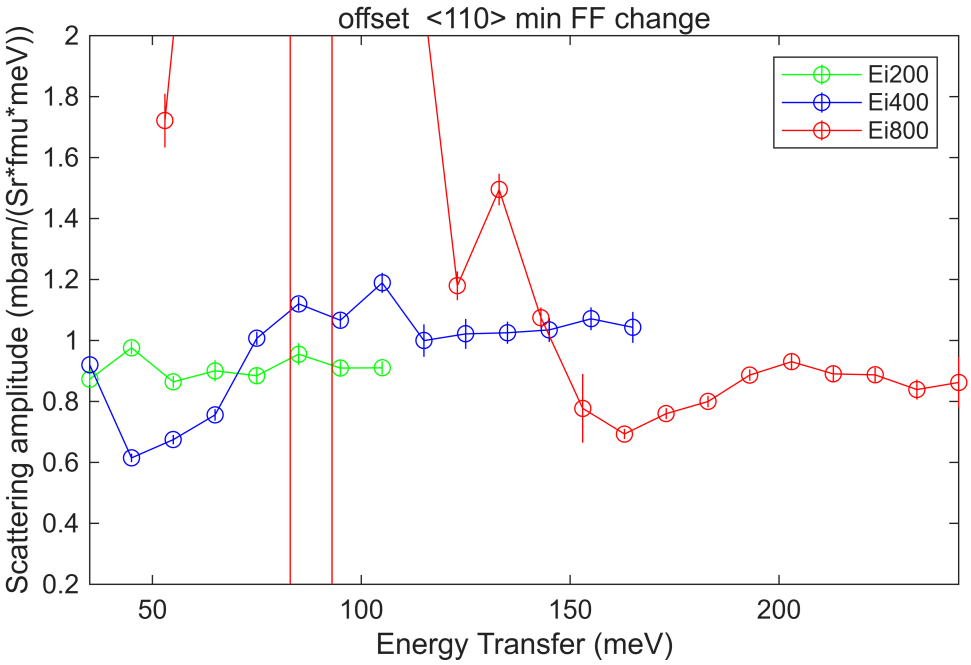
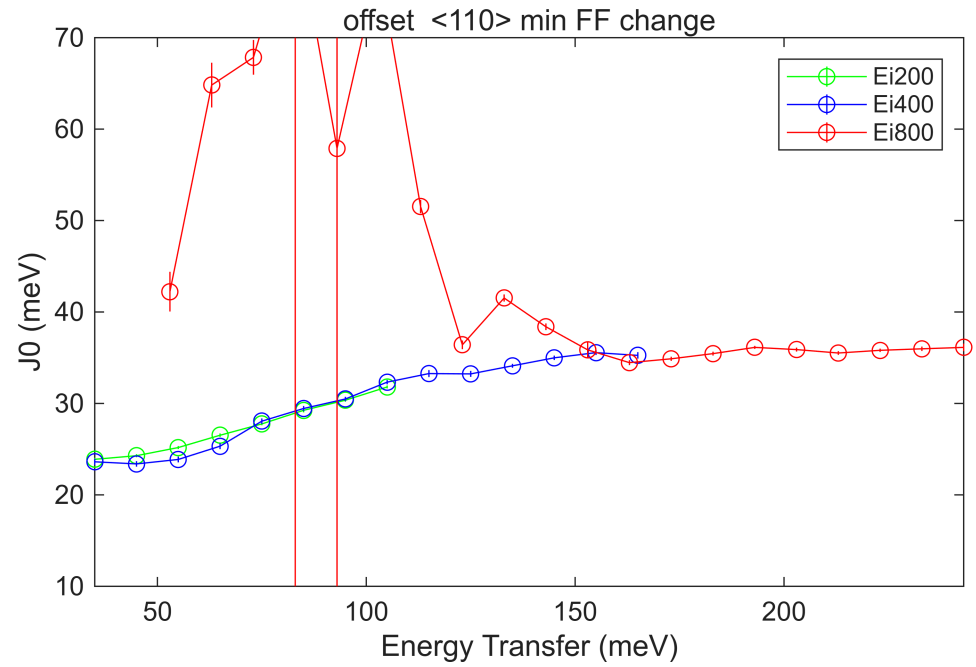
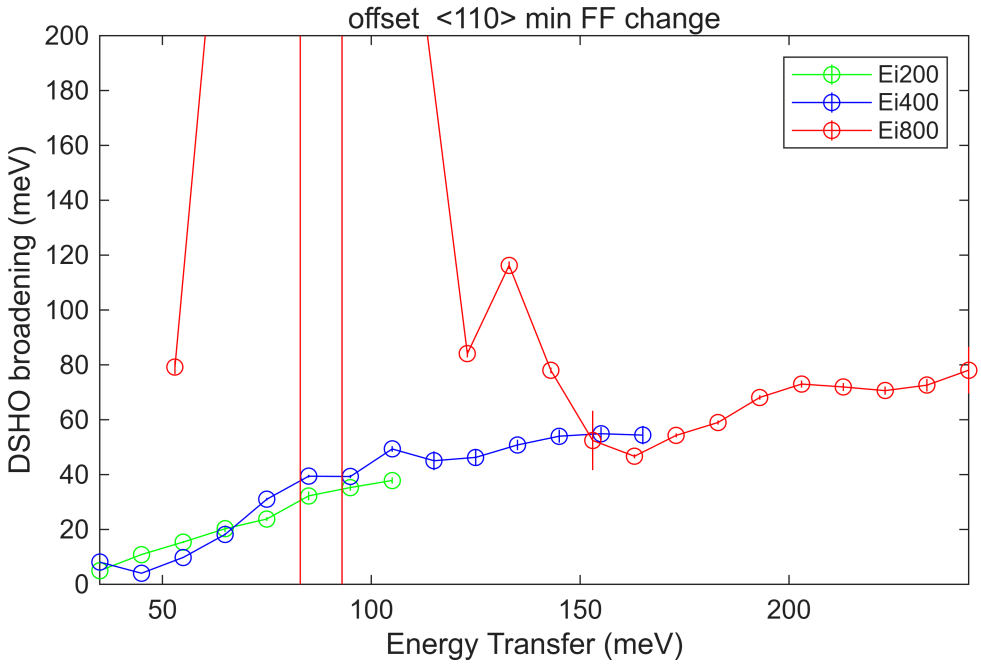
```
%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
[NG_FFCutsFits,FileResName] =
recover_selected_cuts('NG_FFCutsFits','e:\SHARE\Fe\Data\analysis\07_fit_with_mult
icut_allJ\sel_cuts');
fn = fieldnames(NG_FFCutsFits);
save_result = false;
for i=1:numel(fn)
    fldVal  = NG_FFCutsFits.(fn{i});
    if iscell(fldVal) && isscalar(fldVal) && isa(fldVal{1},'sqw')
        save_result = true;
        NG_FFCutsFits.(fn{i}) = fldVal{1};
    end
end
if save_result
    save(FileResName,'NG_FFCutsFits','-v7.3');
end


fit_res_field = 'all_fit_par';
separate_fig  = false; % if true, detach 1D figures from live script
```

```
[S_op110,J_op110,G_op110,Sq_op110,proj110minFF]=
display_EiDep_blocks(NG_FFCutsFits, ...
    'off110NGminFF',' <110> min FF change',fit_res_field,separate_fig);
```

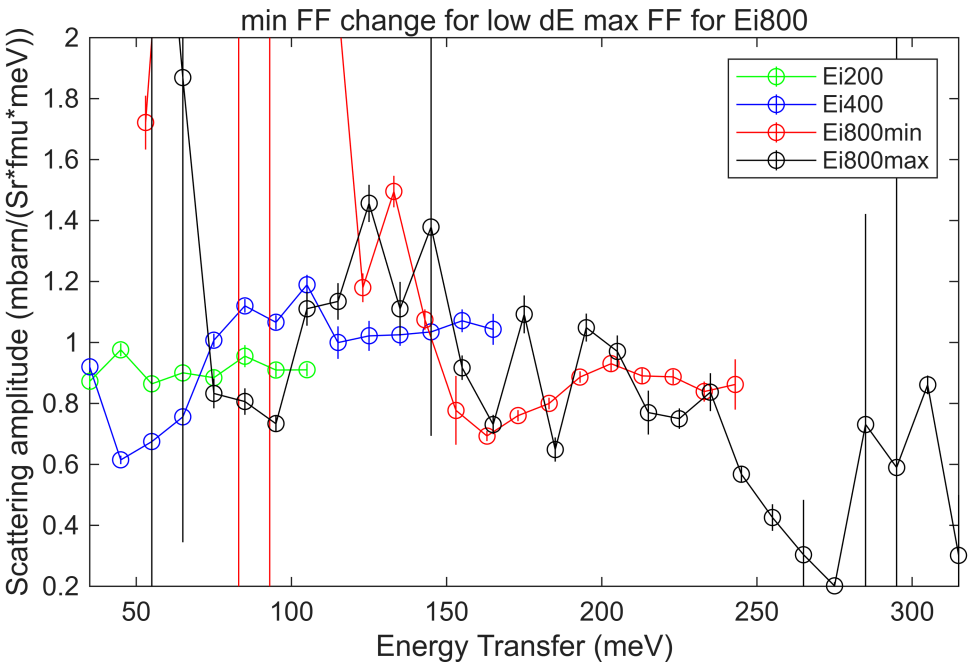


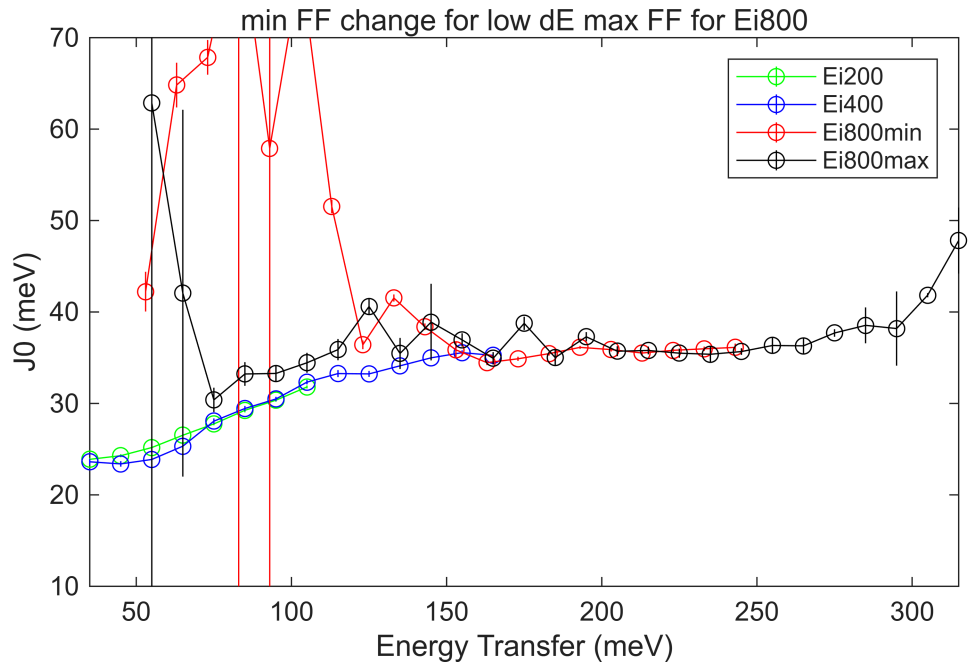


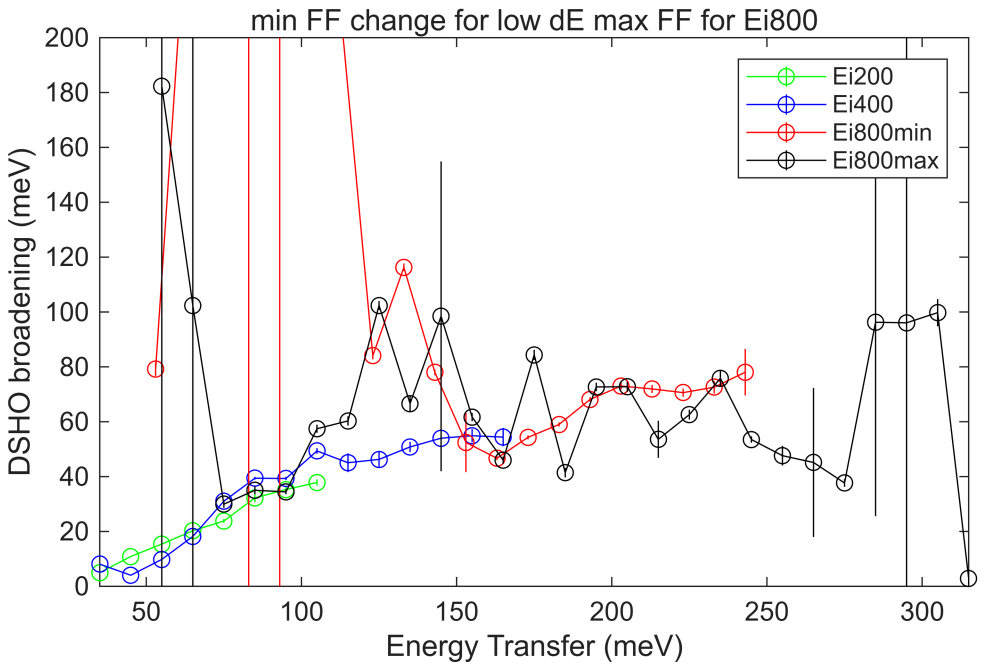



```
[S_maEi800,J_maEi800,Gam_maEi800,S_maEi800q] = extract_fit_par( ...
    NG_FFCutsFits.(['Ei800_off110NGmaxFF_',fit_res_field]));
proj200maxFF = NG_FFCutsFits.Ei200_off200NGmaxFF.data.proj;

plot_fp(separate_fig,'min FF change for low dE max FF for Ei800', ...
    [S_op110,S_maEi800], ...
    [J_op110,J_maEi800], ...
    [G_op110,Gam_maEi800]);
```

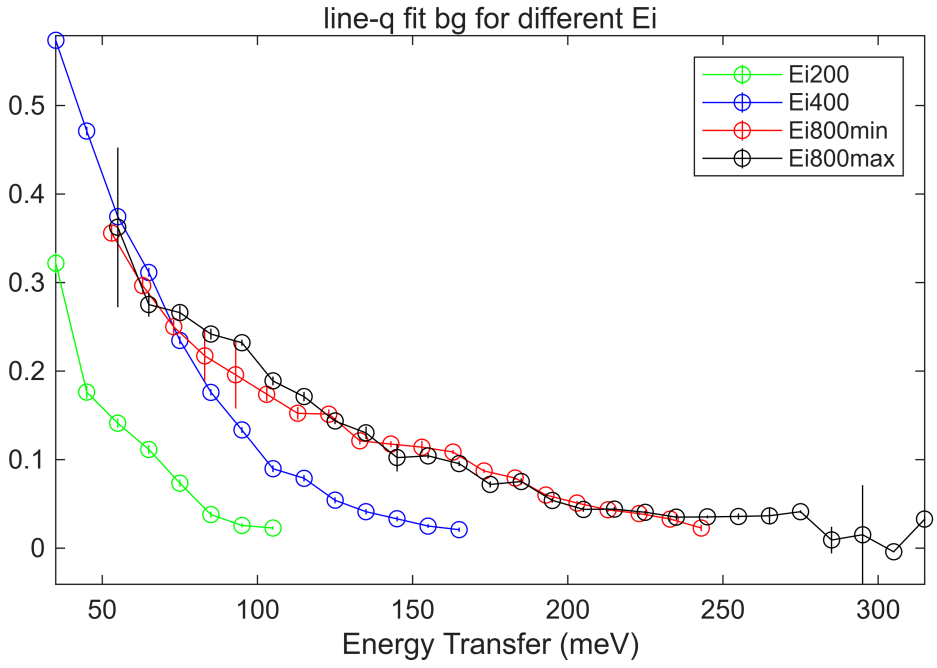

min FF change for low dE max FF for Ei800
Scattering amplitude (mbarn/(Sr*fmu*meV))
Energy Transfer (meV)
Ei200
Ei400
Ei800min
Ei800max


min FF change for low dE max FF for Ei800
J0 (meV)
Energy Transfer (meV)
Ei200
Ei400
Ei800min
Ei800max



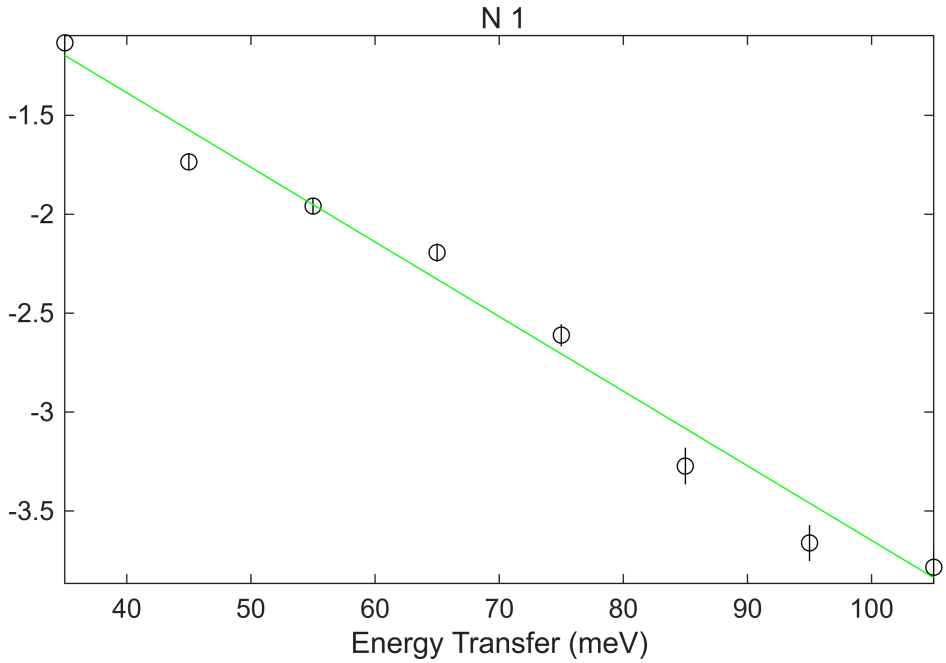

```
bg_dataNGEi200 = extract_bg_par(NG_FFCutsFits.
(['Ei200_off110NGminFF_',fit_res_field]),1);
bg_dataNGEi400 = extract_bg_par(NG_FFCutsFits.
(['Ei400_off110NGminFF_',fit_res_field]),1);
bg_dataNGEi800mi = extract_bg_par(NG_FFCutsFits.
(['Ei800_off110NGminFF_',fit_res_field]),1);
%bg_dataNGEi800ma = extract_bg_par(NG_FFCutsFits.
(['Ei800_off200NGmaxFF_',fit_res_field]),1);
bg_dataNGEi800ma = extract_bg_par(NG_FFCutsFits.
(['Ei800_off110NGmaxFF_',fit_res_field]),1);

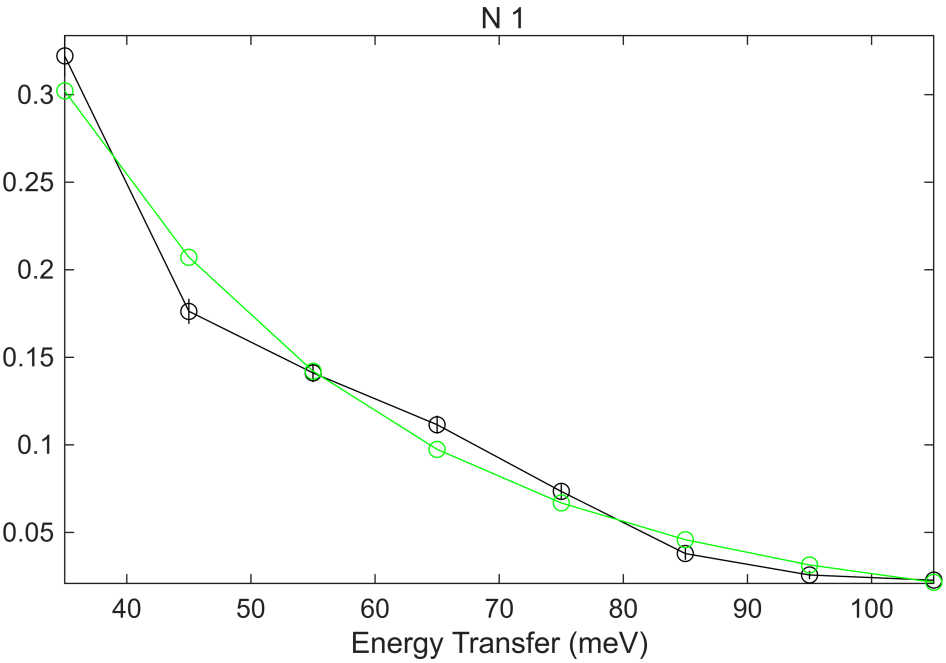
plot_fp(separate_fig,'line-q fit bg for different Ei',[], ...
    [bg_dataNGEi200,bg_dataNGEi400,bg_dataNGEi800mi,bg_dataNGEi800ma])
```



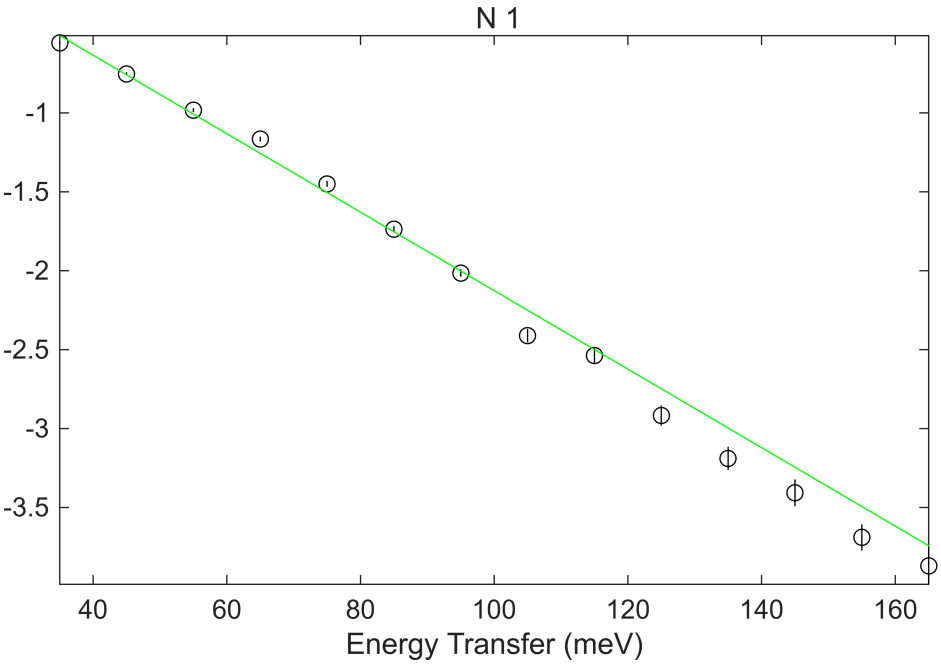
```
%
```

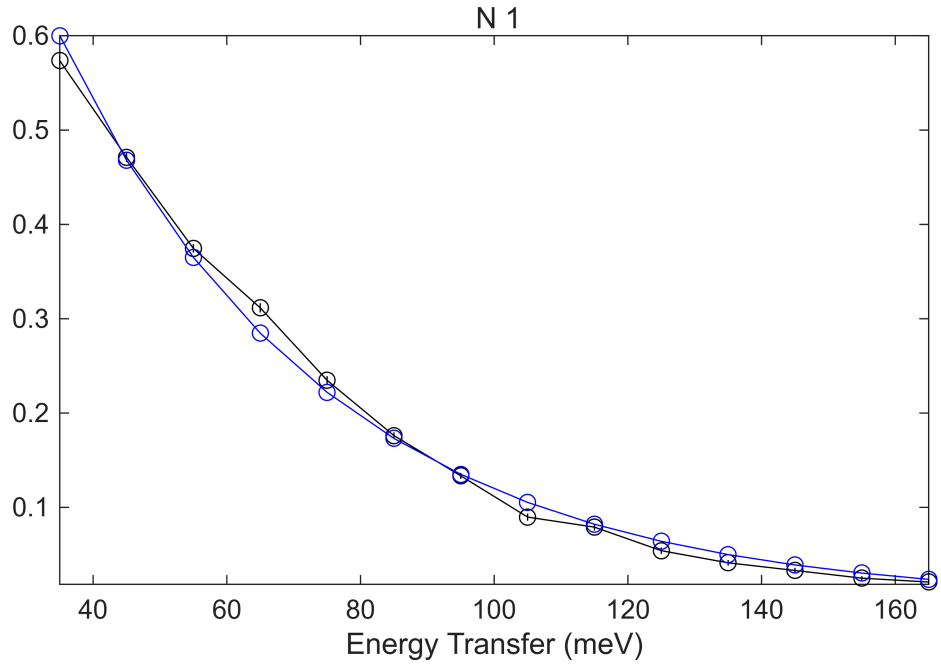
```
[~,bg_dataNGEi200] = fit_IXDsBg_helper(bg_dataNGEi200,[],true,'g');
```



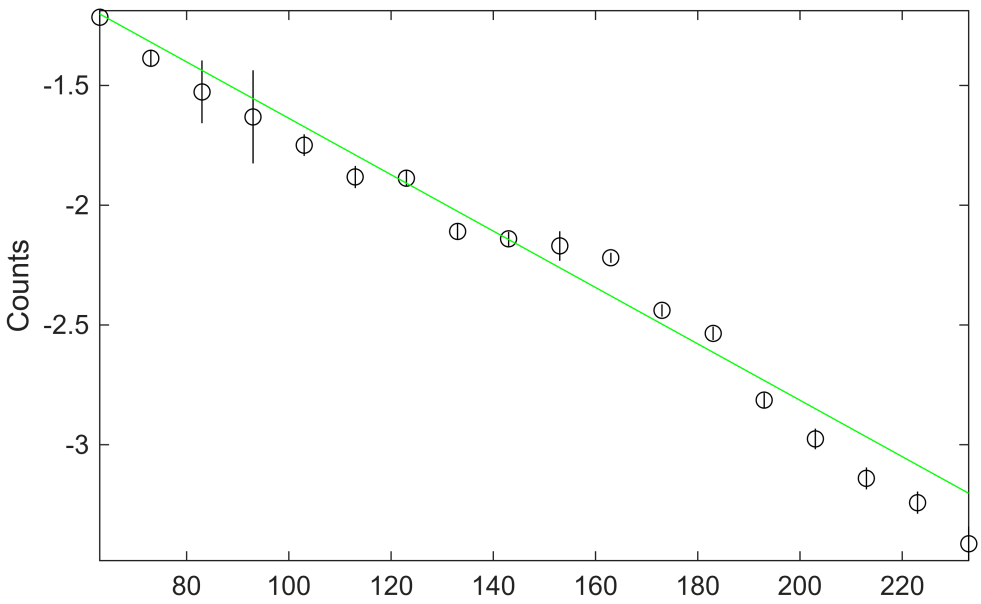


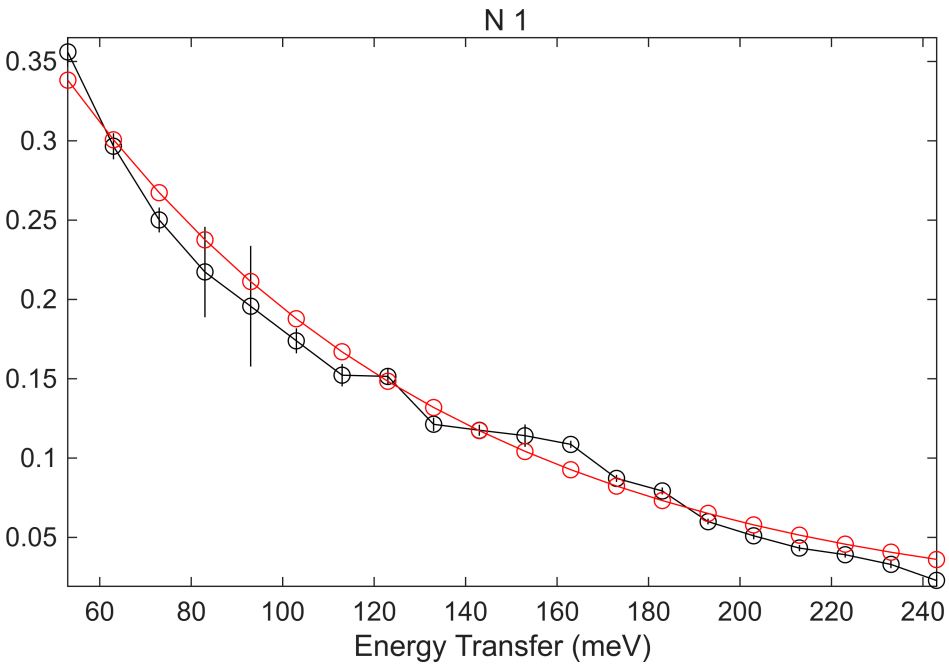

```
[~,bg_dataNGEi400] = fit_IXDsBg_helper(bg_dataNGEi400,[],true,'b');
```



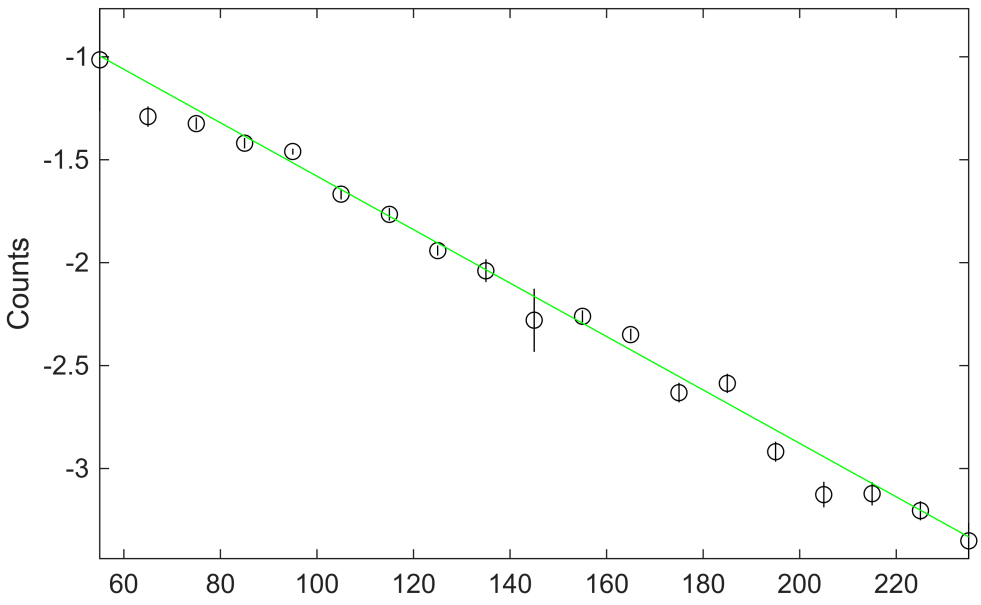


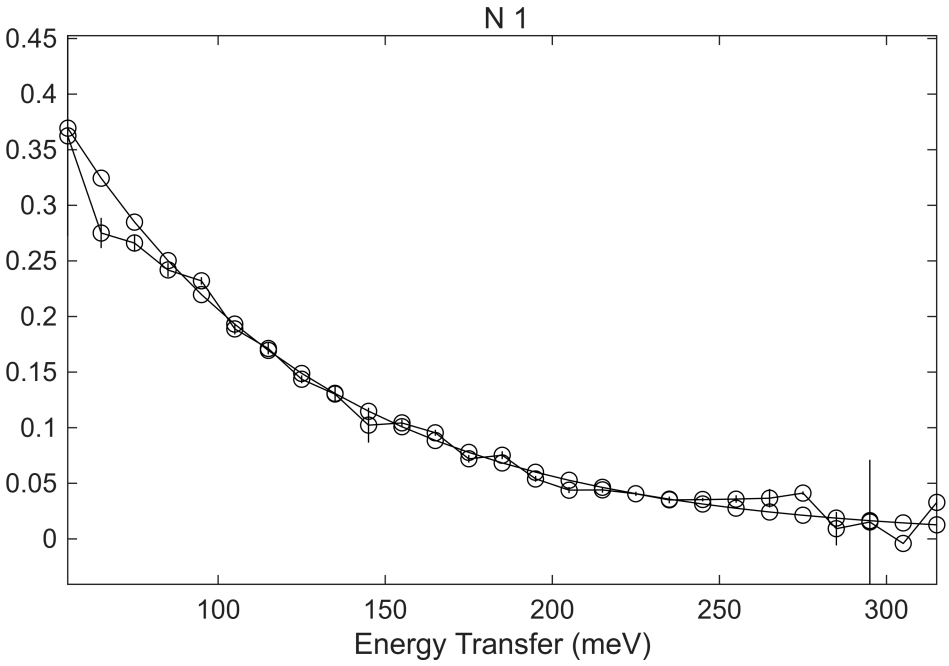

```
[~,bg_dataNGEi800mi] = fit_IXDsBg_helper(bg_dataNGEi800mi,[55,235],true,'r');
```



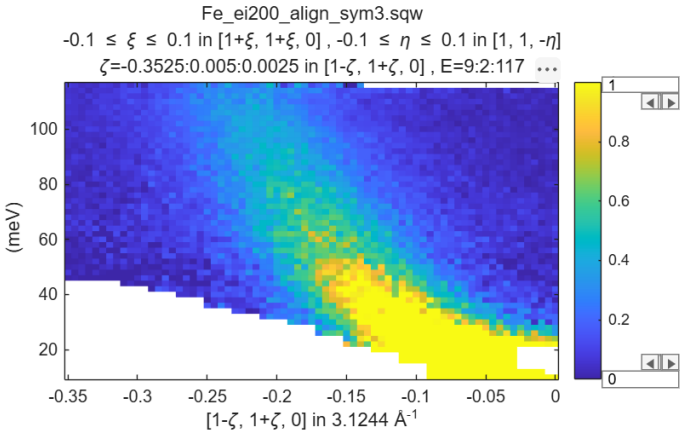



```
[~,bg_dataNGEi800ma] = fit_IXDsBg_helper(bg_dataNGEi800ma,[55,235],true,'k');
```

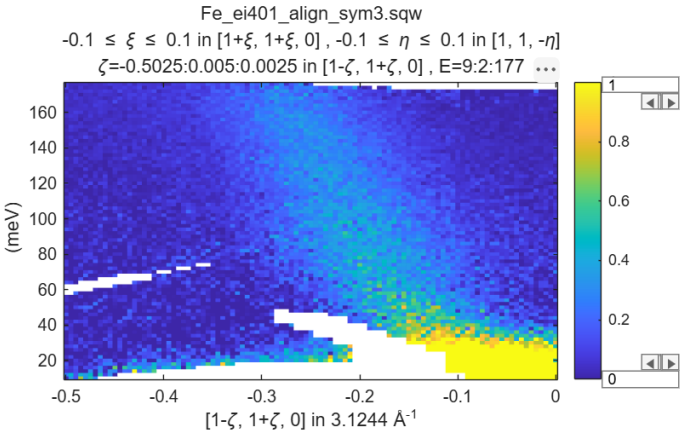




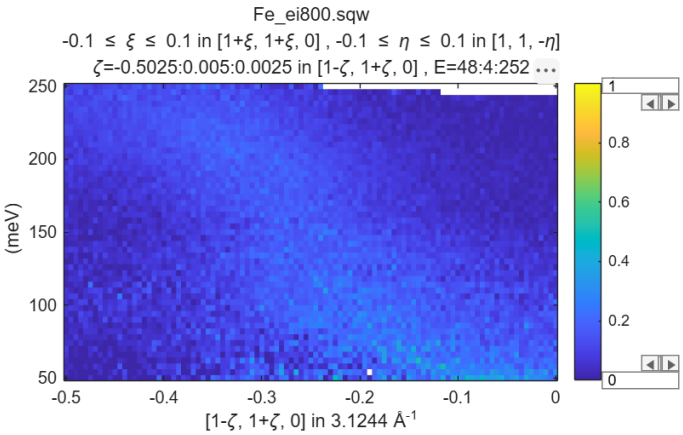

```
% remove calculated backgrounds
spp.Ei200_off110NGminNbg = NG_FFCutsFits.Ei200_off110NGminFF - ...
    sqw_eval(NG_FFCutsFits.Ei200_off110NGminFF,@linterp_bg,bg_dataNGEi200);
spp.Ei200_off110NGminNbg = cut(spp.Ei200_off110NGminNbg, ...
    [-0.35,0.005,0],[],SymopReflection('normvec',[-1,1,0],'offset',[1,1,0]));
plot(spp.Ei200_off110NGminNbg); lz 0 1; keep_figure
```




```
spp.Ei400_off110NGminNbg = NG_FFCutsFits.Ei400_off110NGminFF - ...
    sqw_eval(NG_FFCutsFits.Ei400_off110NGminFF,@linterp_bg,bg_dataNGEi400);
spp.Ei400_off110NGminNbg = cut(spp.Ei400_off110NGminNbg,[-0.5,0.005,0],
[],SymopReflection('normvec',[-1,1,0],'offset',[1,1,0]));
plot(spp.Ei400_off110NGminNbg); lz 0 1; keep_figure
```

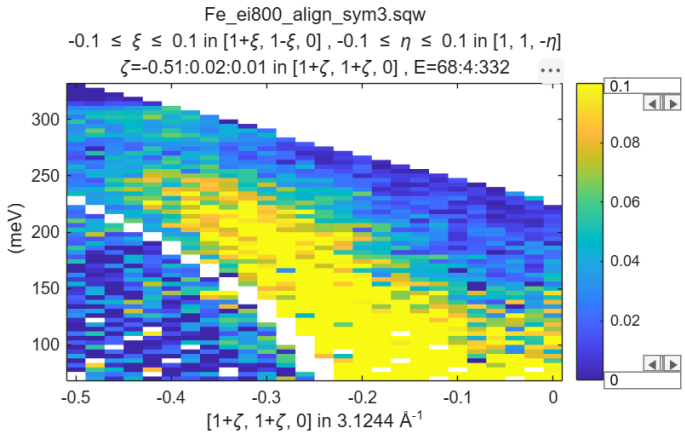



```
spp.Ei800_off110NGminNbg = NG_FFCutsFits.Ei800_off110NGminFF - ...
    sqw_eval(NG_FFCutsFits.Ei800_off110NGminFF,@linterp_bg,bg_dataNGEi800mi);
spp.Ei800_off110NGminNbg = cut(spp.Ei800_off110NGminNbg, ...
    [-0.5,0.005,0],[],SymopReflection('normvec',[-1,1,0],'offset',[1,1,0]));
plot(spp.Ei800_off110NGminNbg); lz 0 1; keep_figure
```



```
%spp.Ei800_off200NGmaxNbg = NG_FFCutsFits.Ei800_off200NGmaxFF - ...
%    sqw_eval(NG_FFCutsFits.Ei800_off200NGmaxFF,@linterp_bg,bg_dataNGEi800ma);
%spp.Ei800_off200NGmaxNbg = cut(spp.Ei800_off200NGmaxNbg, ...
%    [-0.5,0.005,0],[],SymopReflection('normvec',[1,1,0],'offset',[2,0,0]))
%plot(spp.Ei800_off200NGmaxNbg); lz 0 0.1; keep_figure

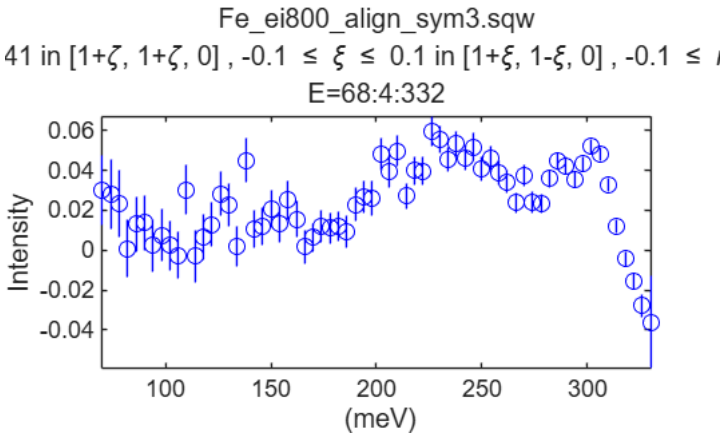
spp.Ei800_off110NGmaxNbg = NG_FFCutsFits.Ei800_off110NGmaxFF - ...
    sqw_eval(NG_FFCutsFits.Ei800_off110NGmaxFF,@linterp_bg,bg_dataNGEi800ma);
spp.Ei800_off110NGmaxNbg = cut(spp.Ei800_off110NGmaxNbg, ...
    [-0.5,0.02,0],[],SymopReflection('normvec',[1,1,0],'offset',[2,0,0]));
plot(spp.Ei800_off110NGmaxNbg); lz 0 0.1; keep_figure
```




```
q0=-0.46
```

```
q0 =
-0.4600
```

```
%q0=-0.475
%w1ei800dE = cut(spp.Ei800_off200NGmaxNbg,[q0-0.025,q0+0.025],[]);
w1ei800dE = cut(spp.Ei800_off110NGmaxNbg,[q0-0.05,q0+0.05],[]);
plot(w1ei800dE)
```

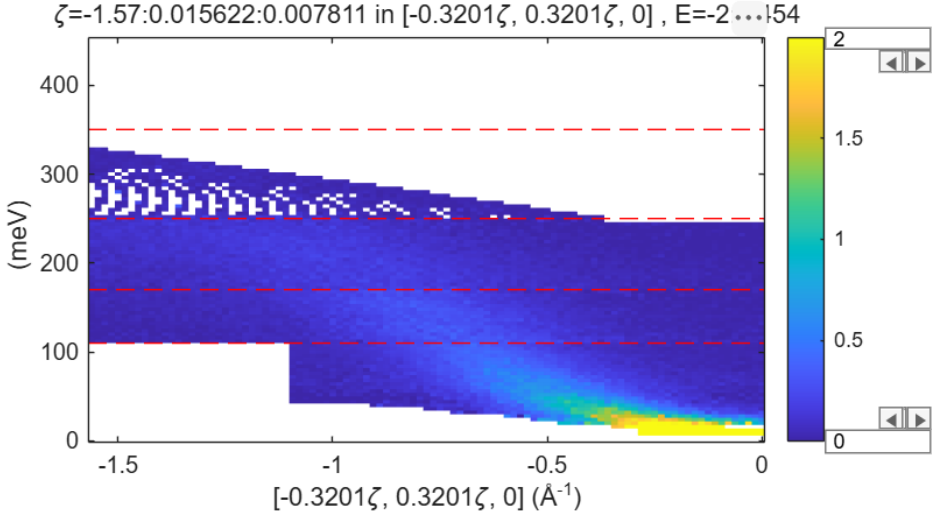



```
cuts_rangesNG = containers.Map();
cuts_rangesNG('Ei200_off110NGminNbg')  = [0;110];
cuts_rangesNG('Ei400_off110NGminNbg') = [110;170];
```

```
cuts_rangesNG('Ei800_off110NGminNbg') = [170;250];
%cuts_rangesNG('Ei800_off200NGmaxNbg') = [250;390];
cuts_rangesNG('Ei800_off110NGmaxNbg') = [250;350];

[w2,fh] = build_disp_in_dir_with_ds(spp,cuts_rangesNG,[-0.5,0.005,0],false);
```

```
*** Finished 1of#4 chunks in 0.84366 sec performing sqw_op_bin_pixels
...
*** Completed sqw_op_bin_pixels using 4 pages in  1.894 sec.
*** Resulting object is backed by file:
                    Fe_ei200_align_sym3.sqw
4 ≤ ξ ≤ 0.31244 in [0.3201ξ, 0.3201ξ, 0] , -0.22093 ≤ η ≤ 0.22093 in [0, 0, -0.4526
      ζ=-1.57:0.015622:0.007811 in [-0.3201ζ, 0.3201ζ, 0] , E=-2...54
```



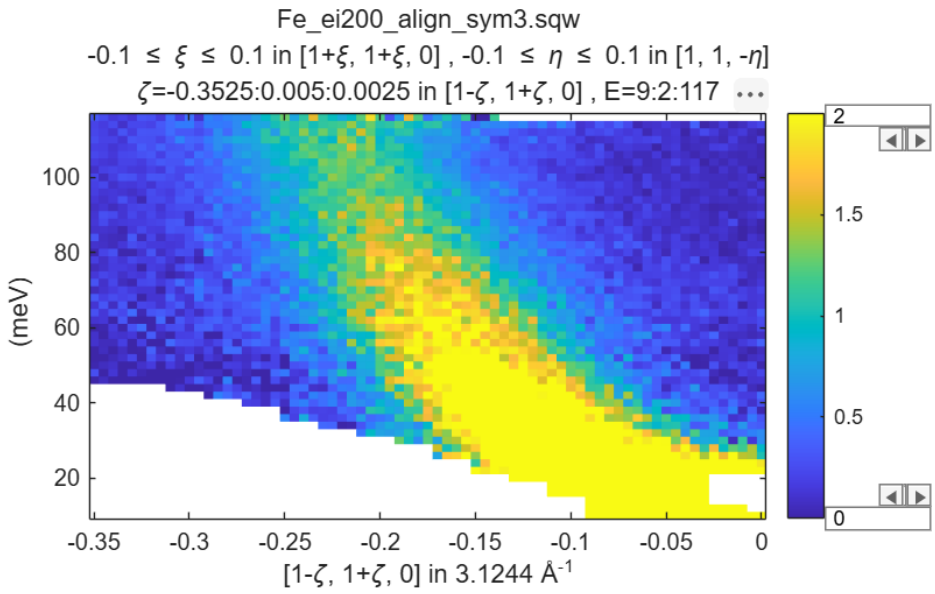

```
data_ranges = cuts_rangesNG.values;
data_ranges  = [data_ranges{:}];
pan_data = struct('data_ranges',data_ranges,'combined_ds',w2);
%save('NG_0p5_0p5_0_dataMFF','pan_data');
fh.UserData = w2.data;
```

```
%plot(spp.Ei800_off200NGmaxNbg); lz 0 0.1;keep_figure;
plot(spp.Ei800_off110NGmaxNbg); lz 0 0.1;keep_figure;
```

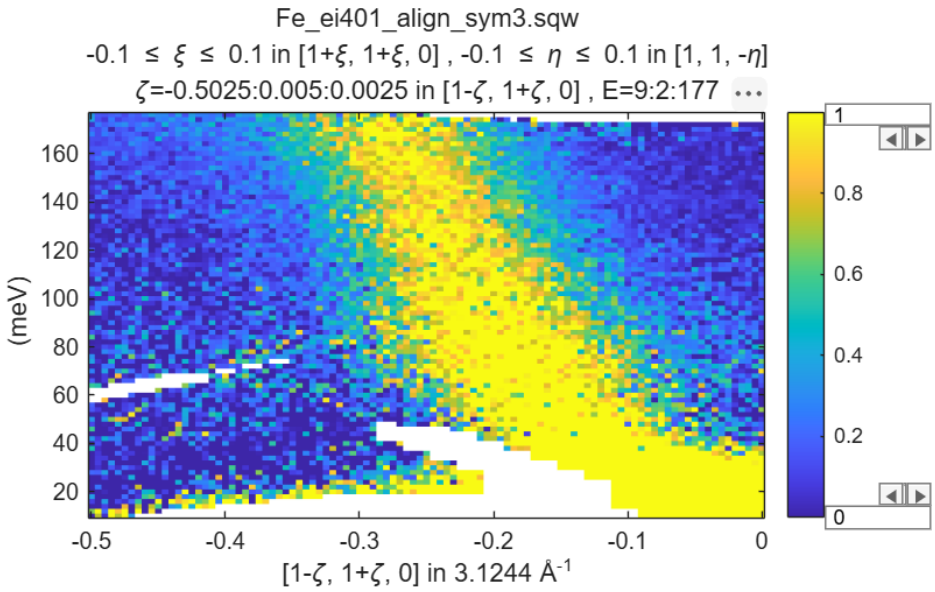
```
Unable to resolve the name 'spp.Ei800_off110NGmaxNbg'.
```

```
%plot(fit_obj); lz 0 0.1;keep_figure;
%[fit_obj,fit_par] = kk.simulate();
```

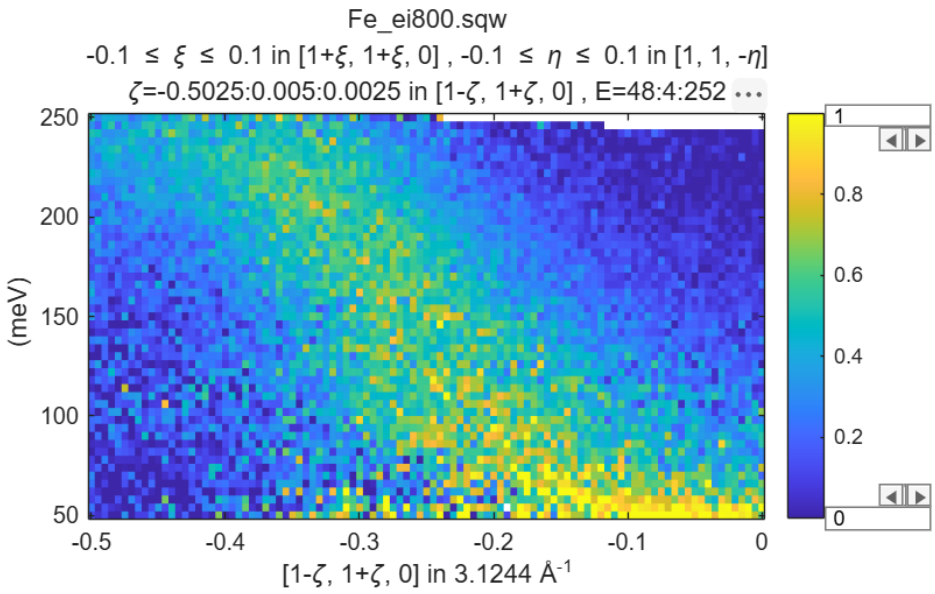
```
if ~exist('magi','var')
    magi = MagneticIons();
end
sppCr.Ei200_off110NGminNbg= magi.correct_mag_ff(spp.Ei200_off110NGminNbg);
plot(sppCr.Ei200_off110NGminNbg); lz 0 2;keep_figure;
```



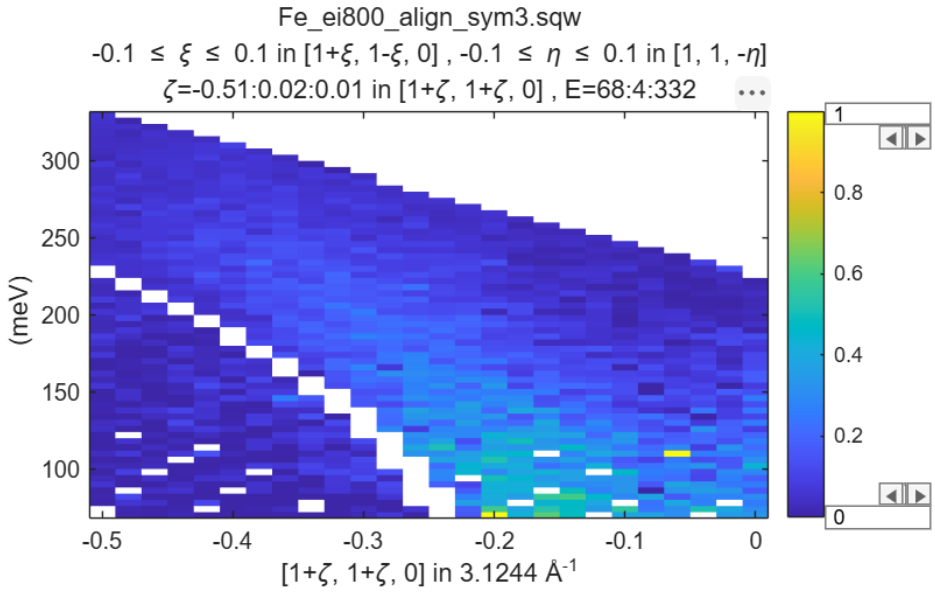

```
sppCr.Ei400_off110NGminNbg = magi.correct_mag_ff(spp.Ei400_off110NGminNbg);
plot(sppCr.Ei400_off110NGminNbg); lz 0 1; keep_figure
```




```
sppCr.Ei800_off110NGminNbg = magi.correct_mag_ff(spp.Ei800_off110NGminNbg);
plot(sppCr.Ei800_off110NGminNbg); lz 0 1; keep_figure
```

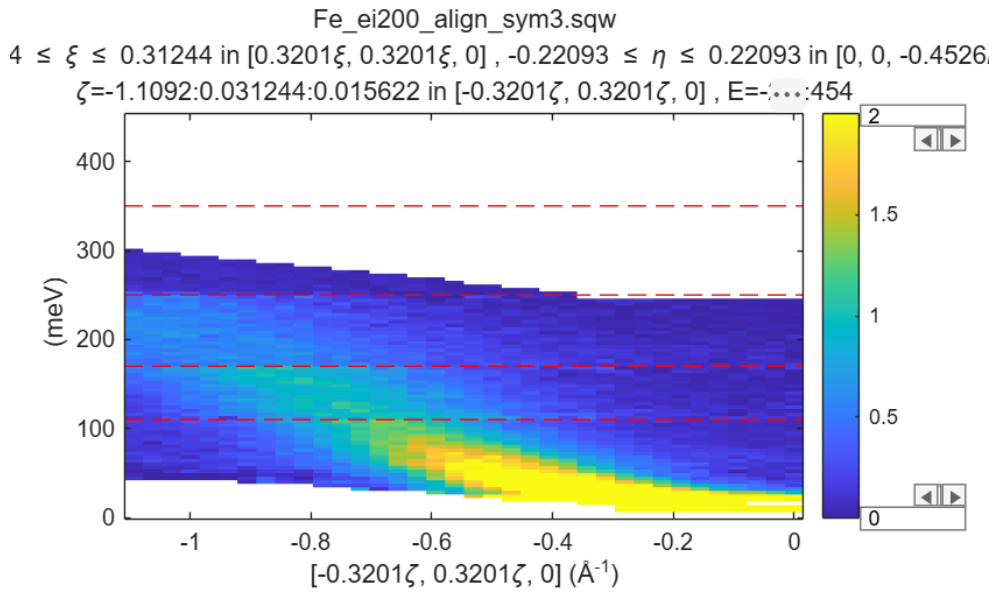



```
%sppCr.Ei800_off200NGmaxNbg = magi.correct_mag_ff(spp.Ei800_off200NGmaxNbg);
%plot(sppCr.Ei800_off200NGmaxNbg); lz 0 1; keep_figure
sppCr.Ei800_off110NGmaxNbg = magi.correct_mag_ff(spp.Ei800_off110NGmaxNbg);
plot(sppCr.Ei800_off110NGmaxNbg); lz 0 1; keep_figure
```




```
[w2,fh] = build_disp_in_dir_with_ds(sppCr,cuts_rangesNG,0.01,0.5,false);
```

```
*** Finished 1of#4 chunks in 0.94678 sec performing sqw_op_bin_pixels
...
*** Completed sqw_op_bin_pixels using 4 pages in 1.9794 sec.
*** Resulting object is backed by file:
```




```
data_ranges = cuts_rangesNG.values;
data_ranges  = [data_ranges{:}];
pan_data = struct('data_ranges',data_ranges,'combined_ds',w2);
save('NG_0p5_0p5_0_dataMFF','pan_data');
```

```
gamma=10;Seff0=0.7917;
%J0=39; J1=35; % GP
J0 = 52; J1 = 20; %  NG
correct_ff = 1;
gap = 0;    %
init_fg_param = [correct_ff,8,gamma,Seff0,  0,   J0, J1,  0,  0,  0];
%init_fg_params = [coffect_ff,T,gamma,Seff, gap, J0, J1, J2, J3, J4];
free_sw_param  =  [0          0, 1   ,1   , 0,    1, 1,   0,  0,  0];


keys = cuts_rangesNG.keys();
n_cuts = 0;
cut_ranges = struct();
for keyIndex = 1:numel(keys)
    currentKey = keys{keyIndex};
    range = cuts_rangesNG(currentKey);
    min_range = 30;
    if contains(currentKey,'Ei400')
        min_range = 40;
    end
    if contains(currentKey,'Ei800')
        min_range = 70;
    end
    if range(1)<min_range
        range(1) = min_range;
    end
    cut_ranges.(currentKey) = range;
    n_cuts = n_cuts + floor(range(2)-range(1))/10;
end
```

```
cuts = cell(1,n_cuts);
cutIdx = 0;
en = zeros(1,n_cuts);
for keyIndex = 1:numel(keys)
    currentKey = keys{keyIndex};
    range = cut_ranges.(currentKey);
    ncc = floor(range(2)-range(1))/10 - 1;
    for i=1:ncc
        cutIdx =cutIdx+1;
        cuts{cutIdx} = cut(spp.(currentKey),[],[range(1)+
(i-1)*10,range(1)+i*10]);
        en(cutIdx) = range(1)+5+10*(i-1);
        %plot(cuts{cutIdx});
    end
end
cuts = cuts(1:cutIdx);
```

```
kk = tobyfit(cuts{:});
kk = kk.set_local_foreground;
kk = kk.set_fun(@sqw_iron);

hkl_proj = spp.Ei800_off200NGmaxNbg.data.proj;
kk = kk.set_pin({init_fg_param,hkl_proj});
kk = kk.set_free(free_sw_param);

kk = kk.set_bind({{6,[6,1]},{7,[7,1]}});

%kk = kk.set_bfun (@linear_bg1D); % set_bfun sets the background functions
%
%bg_param = init_bg_param(valid);
%kk = kk.set_bpin (bg_param);  % initial background constant and gradient
%kk = kk.set_bfree (bfree);

kk = kk.set_options('list',2);

[fit_obj,fit_par] = kk.fit();
```

```
Warning: Dataset 20:2 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 1.9802
Warning: Dataset 22:3 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 2.9703
Warning: Dataset 24:1 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 0.9901
Warning: Dataset 25:1 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 0.9901
Warning: Dataset 26:1 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 0.9901
Warning: Dataset 27:3 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 2.9703
Warning: Dataset 28:4 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 3.9604
Warning: Dataset 29:3 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 2.9703
Warning: Dataset 31:3 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 2.9703
```

```
Warning: Dataset 32:1 points with zero, negative or undefined (infinite or NaN) error bars have been
removed from fit, which is 0.9901
------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------
Starting point
------------------
 Total time = 52.0804s    Reduced Chi^2 = 65.1865
 Free parameter values:
           10         0.7917             52             20             10        0.7917             10
------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 308.1365s    Reduced Chi^2 = 28.4569      Levenberg-Marquardt = 0.1
 Free parameter values:
         62.8          1.751           48.1             20          47.83         1.146          34.82
------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 560.4997s    Reduced Chi^2 = 10.4019      Levenberg-Marquardt = 0.01
 Free parameter values:
        134.3          3.231          45.74             20            160         2.913          156.9
------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 814.4295s    Reduced Chi^2 = 7.554      Levenberg-Marquardt = 0.001
 Free parameter values:
        31.21          1.936           36.4             20          30.14         1.609          52.02
------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 1064.9219s    Reduced Chi^2 = 18.1239      Levenberg-Marquardt = 0.0001
 Free parameter values:
        40.08          1.726          30.84             20           41.7         1.451          49.53
 Total time = 1116.5597s    Reduced Chi^2 = 18.0426      Levenberg-Marquardt = 0.001
 Free parameter values:
        40.28          1.733          30.91             20           41.9         1.456          49.86
 Total time = 1168.159s    Reduced Chi^2 = 17.9074      Levenberg-Marquardt = 0.01
 Free parameter values:
         41.8          1.784          31.53             20          43.44         1.495          52.46
 Total time = 1219.4847s    Reduced Chi^2 = 3.76      Levenberg-Marquardt = 1
 Free parameter values:
        41.94          1.837          35.86             20          42.62         1.508           61.4
------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 1476.4097s    Reduced Chi^2 = 2.9607      Levenberg-Marquardt = 0.1
 Free parameter values:
         50.9          1.922          33.87             20          55.98         1.636          62.23
------------------------------------------------------------------------------
Iteration = 6
------------------
 Total time = 1734.5507s    Reduced Chi^2 = 3.1279      Levenberg-Marquardt = 0.01
 Free parameter values:
         47.9          1.863          33.91             20          53.09         1.593          58.52
 Total time = 1785.7873s    Reduced Chi^2 = 2.9138      Levenberg-Marquardt = 0.1
 Free parameter values:
        48.28          1.872          33.93             20          53.43         1.599          58.93
------------------------------------------------------------------------------
Iteration = 7
------------------
 Total time = 2032.9775s    Reduced Chi^2 = 6.1632      Levenberg-Marquardt = 0.01
 Free parameter values:
        48.14          1.865          33.78             20          53.11         1.591          58.43
 Total time = 2081.5471s    Reduced Chi^2 = 2.9096      Levenberg-Marquardt = 0.1
 Free parameter values:
        48.24          1.868           33.8             20          53.24         1.593          58.58
------------------------------------------------------------------------------
Iteration = 8
```

```
------------------
 Total time = 2336.8084s    Reduced Chi^2 = 2.8944      Levenberg-Marquardt = 0.01
 Free parameter values:
         47.8          1.859          33.77             20          52.82         1.588          58.11
-------------------------------------------------------------------------------
Iteration = 9
------------------
 Total time = 2592.9528s    Reduced Chi^2 = 15.7593      Levenberg-Marquardt = 0.001
 Free parameter values:
         47.8          1.859          33.75             20          52.78         1.586          58.05
 Total time = 2643.5632s    Reduced Chi^2 = 2.9714      Levenberg-Marquardt = 0.01
 Free parameter values:
         47.8          1.859          33.75             20          52.78         1.586          58.06
 Total time = 2692.2046s    Reduced Chi^2 = 2.8973      Levenberg-Marquardt = 0.1
 Free parameter values:
        47.81          1.859          33.75             20           52.8         1.587          58.07
 Total time = 2742.2073s    Reduced Chi^2 = 2.8911      Levenberg-Marquardt = 10
 Free parameter values:
        47.81          1.859          33.76             20          52.82         1.587           58.1
-------------------------------------------------------------------------------
Iteration = 10
------------------
 Total time = 2998.2307s    Reduced Chi^2 = 2.8965      Levenberg-Marquardt = 1
 Free parameter values:
        47.78          1.859          33.76             20          52.75         1.587          58.09
 Total time = 3049.4974s    Reduced Chi^2 = 2.893      Levenberg-Marquardt = 10
 Free parameter values:
         47.8          1.859          33.76             20           52.8         1.587           58.1
 Total time = 3100.5366s    Reduced Chi^2 = 2.8919      Levenberg-Marquardt = 100
 Free parameter values:
         47.8          1.859          33.76             20          52.81         1.587           58.1
 Total time = 3151.9746s    Reduced Chi^2 = 2.895      Levenberg-Marquardt = 10000
 Free parameter values:
        47.81          1.859          33.76             20          52.82         1.587           58.1
 Total time = 3203.228s    Reduced Chi^2 = 2.8913      Levenberg-Marquardt = 1000000
 Free parameter values:
        47.81          1.859          33.76             20          52.82         1.587           58.1
 Total time = 3254.7184s    Reduced Chi^2 = 2.8942      Levenberg-Marquardt = 100000000
 Free parameter values:
        47.81          1.859          33.76             20          52.82         1.587           58.1
 *** No improvement in chi-squared over previous iteration ***
 Total time = 3254.7436s    Reduced Chi^2 = 2.8911
 Free parameter values:
        47.81          1.859          33.76             20          52.82         1.587           58.1
-------------------------------------------------------------------------------
Fit converged:

Parameter values (free parameters with error estimates):
---------------------------------------------------------

Foreground parameter values:
----------------------------
  Function 1:
    1              1
    2              8
    3          47.81   +/-   Inf
    4          1.859   +/-   Inf
    5              0
    6          33.76   +/-   Inf
    7             20   +/-   Inf
    8              0
    9              0
   10              0

  Function 2:
    1              1
    2              8
    3          52.82   +/-   Inf
```

```
  4          1.587   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 3:
  1              1
  2              8
  3           58.1   +/-   Inf
  4           1.49   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 4:
  1              1
  2              8
  3          54.99   +/-   Inf
  4          1.294   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 5:
  1              1
  2              8
  3          54.29   +/-   Inf
  4          1.215   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 6:
  1              1
  2              8
  3          53.14   +/-   Inf
  4          1.119   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 7:
  1              1
  2              8
  3          49.49   +/-   Inf
  4          1.019   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0
```

```
Function 8:
  1              1
  2              8
  3          49.57   +/-   Inf
  4          1.049   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 9:
  1              1
  2              8
  3          48.16   +/-   Inf
  4          1.025   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 10:
  1              1
  2              8
  3           48.4   +/-   Inf
  4         0.9797   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 11:
  1              1
  2              8
  3          48.08   +/-   Inf
  4         0.9793   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 12:
  1              1
  2              8
  3          47.19   +/-   Inf
  4         0.9564   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 13:
  1              1
  2              8
  3          49.82   +/-   Inf
  4         0.7317   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
```

```
  8              0
  9              0
 10              0

Function 14:
  1              1
  2              8
  3             51   +/-   Inf
  4         0.7493   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 15:
  1              1
  2              8
  3          60.75   +/-   Inf
  4          0.811   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 16:
  1              1
  2              8
  3          60.16   +/-   Inf
  4         0.8066   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 17:
  1              1
  2              8
  3          59.24   +/-   Inf
  4           0.77   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 18:
  1              1
  2              8
  3           66.9   +/-   Inf
  4         0.7802   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 19:
  1              1
  2              8
  3          76.45   +/-   Inf
```

```
  4         0.7901   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 20:
  1              1
  2              8
  3          150.1   +/-   Inf
  4           1.67   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 21:
  1              1
  2              8
  3           86.6   +/-   Inf
  4         0.8219   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 22:
  1              1
  2              8
  3          59.32   +/-   Inf
  4          0.469   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 23:
  1              1
  2              8
  3          33.76   +/-   Inf
  4         0.3699   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 24:
  1              1
  2              8
  3          298.2   +/-   Inf
  4          2.142   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0
```

```
Function 25:
  1              1
  2              8
  3          196.6   +/-   Inf
  4          1.341   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 26:
  1              1
  2              8
  3          222.5   +/-   Inf
  4         0.9923   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 27:
  1              1
  2              8
  3          214.4   +/-   Inf
  4          1.311   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 28:
  1              1
  2              8
  3           2500   +/-   Inf
  4          11.15   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 29:
  1              1
  2              8
  3          268.4   +/-   Inf
  4          2.006   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
  8              0
  9              0
 10              0

Function 30:
  1              1
  2              8
  3         -187.3   +/-   Inf
  4           1.42   +/-   Inf
  5              0
  6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
  7             20   +/-   Inf               bound to parameter 7 of foreground 1
```

```
    8              0
    9              0
   10              0

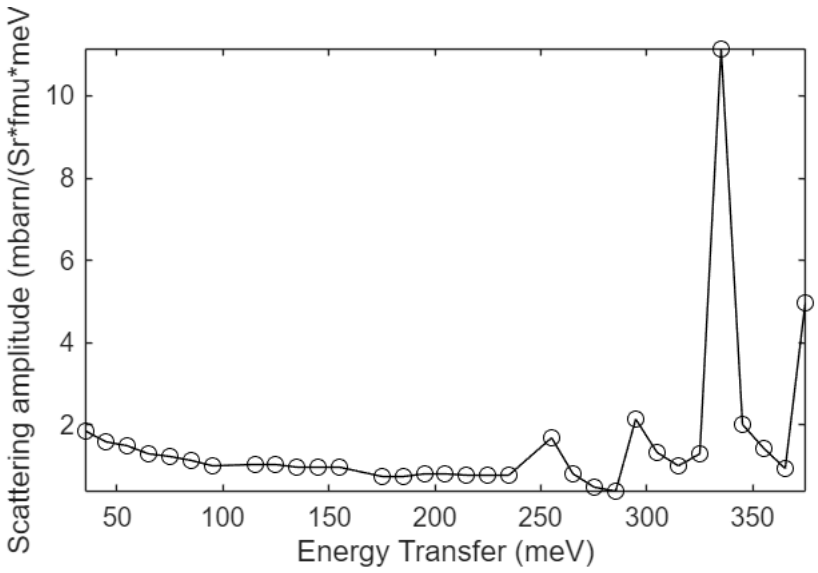
  Function 31:
    1              1
    2              8
    3          114.9   +/-   Inf
    4         0.9534   +/-   Inf
    5              0
    6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
    7             20   +/-   Inf               bound to parameter 7 of foreground 1
    8              0
    9              0
   10              0

  Function 32:
    1              1
    2              8
    3         -4.916   +/-   Inf
    4          4.978   +/-   Inf
    5              0
    6          33.76   +/-   Inf               bound to parameter 6 of foreground 1
    7             20   +/-   Inf               bound to parameter 7 of foreground 1
    8              0
    9              0
   10              0

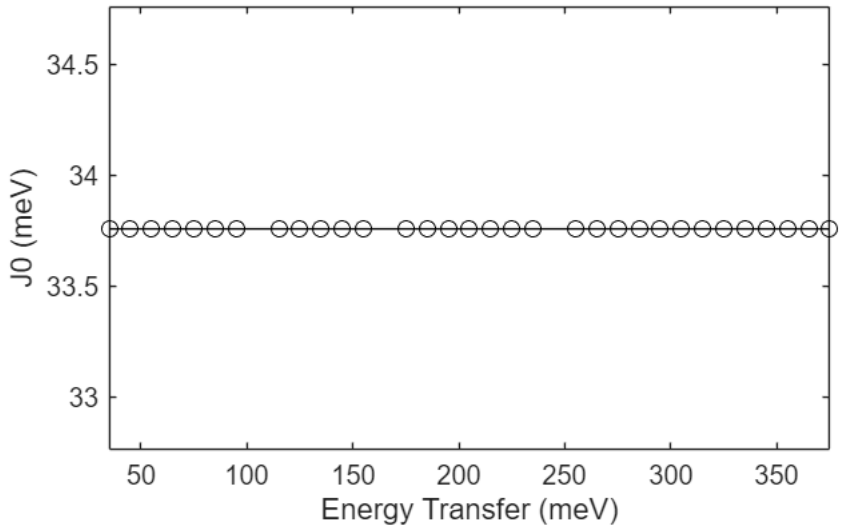

Covariance matrix for free parameters:
--------------------------------------
   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
```

```
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   NaN   N
```

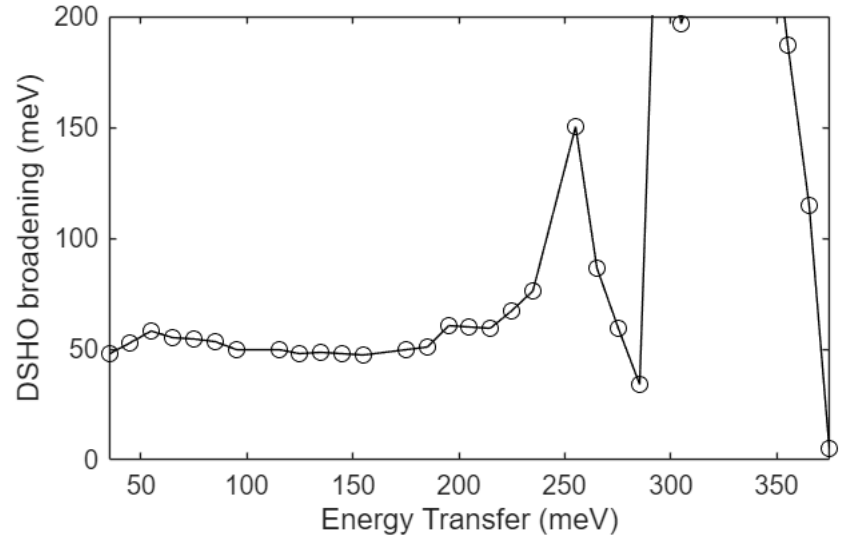
```
[Sar,J0ar,GamAr,J1ar] = extract_multicut_fit_par(fit_par,en);
pd(Sar); keep_figure
```



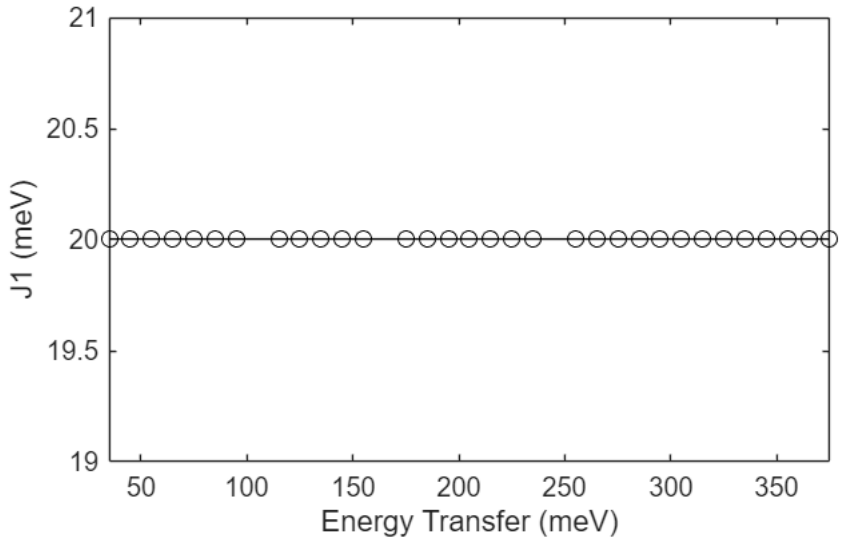

```
pd(J0ar); keep_figure
```




```
pd(GamAr);ly 0 200; keep_figure;
```

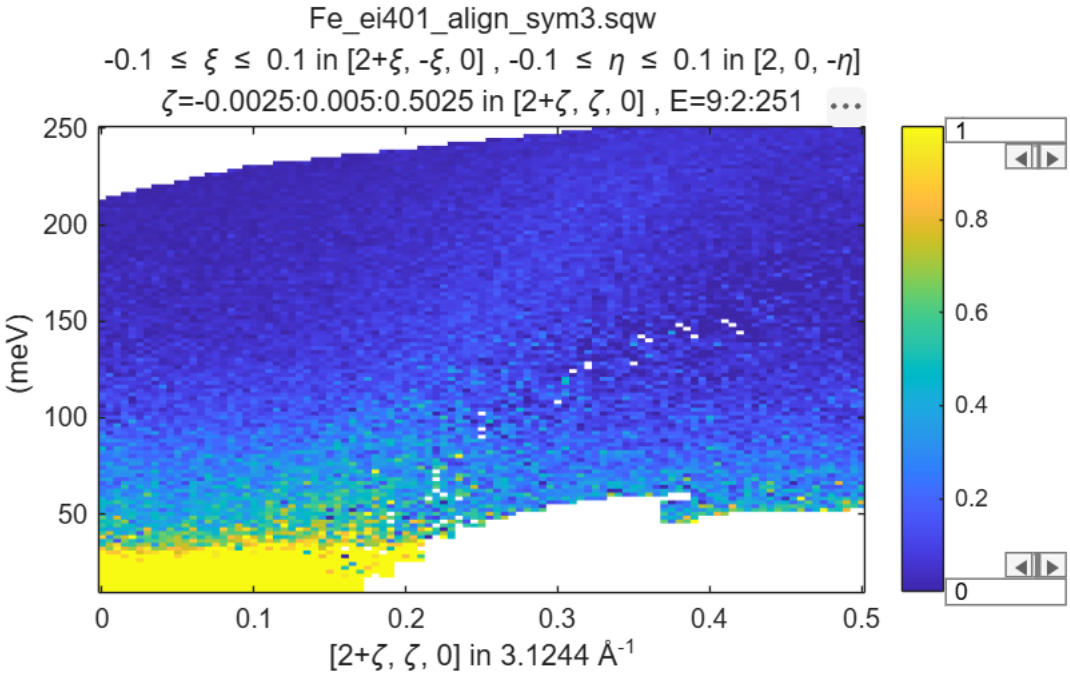



```
pd(J1ar);keep_figure;
```



```
gamma=10;Seff0=0.7917;
%J0=39; J1=35; % GP
J0 = 44; J1 = 20; %  NG
correct_ff = 1;
gap = 0;    %
init_fg_param = [correct_ff,8,gamma,Seff0,  0,   J0, J1,  0,  0,  0];
%init_fg_params = [coffect_ff,T,gamma,Seff, gap, J0, J1, J2, J3, J4];
free_sw_param  =  [0          0, 1   ,1   , 0,    1, 1,   0,  0,  0];
selCut = NG_FFCutsFits.Ei400_off200NGmaxFF;
mi = maps_instrument(401,600,'S');
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
selCut = selCut.set_instrument(mi);
selCut = selCut.set_sample(sample);

plot(selCut); lz 0 1;keep_figure;
```




```
e_sel = [70,100,150,180,210];
cuts = cell(1,numel(e_sel));
for i=1:numel(e_sel)
    cuts{i} = cut(selCut,[],[e_sel(i)-5,e_sel(i)+5]);
    plot(cuts{i});
end
```

```
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for    54101 pixels -- processing data... -----> retained     47282 pixels
--------------------------------------------------------------------------------
Number of points in input object: 1993048
     Fraction of object selected:     2.71% (=     54101 points) ;  Access speed:     0.16GB/sec
     Fraction of object retained:     2.37% (=     47282 points)

**** Total time in cut(sqw)   :      0.1sec Including:
     Data access time fraction:  22.9%; Transf. time frac.:  51.0%; Resulting pix preparation:  24.7%
--------------------------------------------------------------------------------
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for    62310 pixels -- processing data... -----> retained     53689 pixels
--------------------------------------------------------------------------------
Number of points in input object: 1993048
```

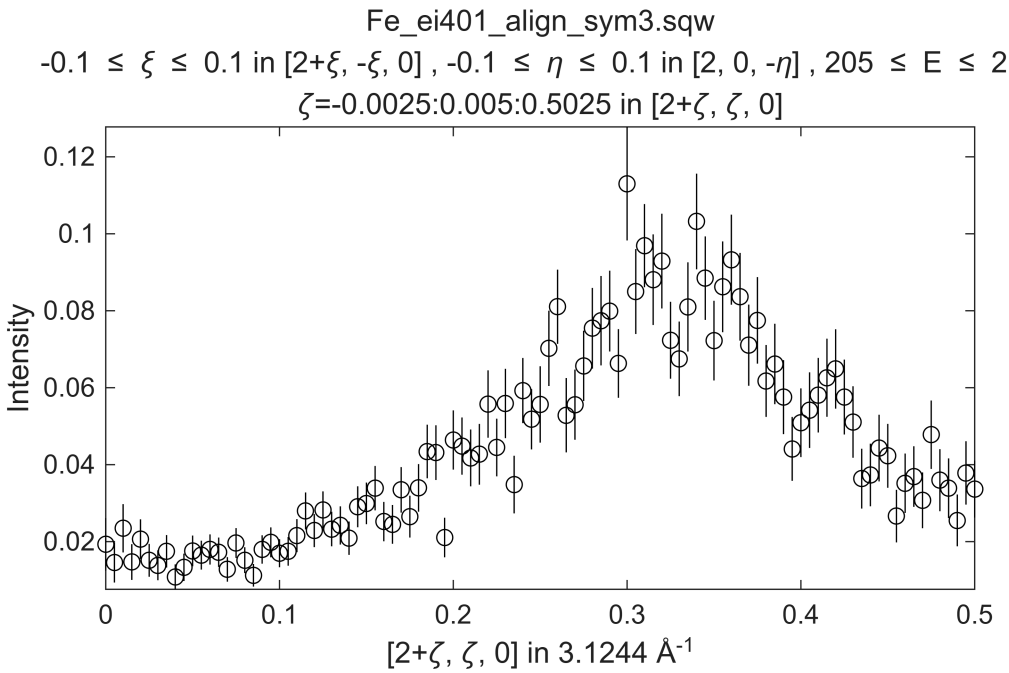
```
     Fraction of object selected:     3.13% (=     62310 points) ;  Access speed:     0.94GB/sec
     Fraction of object retained:     2.69% (=     53689 points)

**** Total time in cut(sqw)   :      0.0sec Including:
     Data access time fraction:  23.8%; Transf. time frac.:  55.0%; Resulting pix preparation:  21.1%
--------------------------------------------------------------------------------
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for    93238 pixels -- processing data... -----> retained     80778 pixels
--------------------------------------------------------------------------------
Number of points in input object: 1993048
     Fraction of object selected:     4.68% (=     93238 points) ;  Access speed:     0.94GB/sec
     Fraction of object retained:     4.05% (=     80778 points)

**** Total time in cut(sqw)   :      0.0sec Including:
     Data access time fraction:  22.6%; Transf. time frac.:  55.9%; Resulting pix preparation:  21.4%
--------------------------------------------------------------------------------
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for   142780 pixels -- processing data... -----> retained    124254 pixels
--------------------------------------------------------------------------------
Number of points in input object: 1993048
     Fraction of object selected:     7.16% (=    142780 points) ;  Access speed:     0.93GB/sec
     Fraction of object retained:     6.23% (=    124254 points)

**** Total time in cut(sqw)   :      0.0sec Including:
     Data access time fraction:  32.1%; Transf. time frac.:  42.2%; Resulting pix preparation:  25.6%
--------------------------------------------------------------------------------
*** Cutting in memory sqw object; returning result in memory; retuning pixels - kept
*** Step 1 of 1; Read data for   281102 pixels -- processing data... -----> retained    245605 pixels
--------------------------------------------------------------------------------
Number of points in input object: 1993048
     Fraction of object selected:    14.10% (=    281102 points) ;  Access speed:     0.91GB/sec
     Fraction of object retained:    12.32% (=    245605 points)

**** Total time in cut(sqw)   :      0.1sec Including:
     Data access time fraction:  37.3%; Transf. time frac.:  32.5%; Resulting pix preparation:  30.1%
--------------------------------------------------------------------------------
```




```
%
```

```
kk = tobyfit(cuts{:});
```

```
kk = kk.set_local_foreground;
kk = kk.set_fun(@sqw_iron);

hkl_proj = selCut.data.proj;
kk = kk.set_pin({init_fg_param,hkl_proj});
kk = kk.set_free(free_sw_param);

kk = kk.set_bind({{6,[6,1]},{7,[7,1]}});

kk = kk.set_bfun (@linear_bg1D); % set_bfun sets the background functions
%
%bg_param = init_bg_param(valid);
kk = kk.set_bpin ([0,0]);  % initial background constant and gradient
kk = kk.set_bfree ([1,0]);

kk = kk.set_options('list',2);

[fit_obj,fit_par] = kk.fit();
```

```
-------------------------------------------------------------------------------
Beginning fit (max 20 iterations)-------------------------------------------------------------------------------------------
Starting point
------------------
 Total time = 6.4039s    Reduced Chi^2 = 21.5734
 Free parameter values:
           10         0.7917             44             20             10         0.7917             10
-------------------------------------------------------------------------------
Iteration = 1
------------------
 Total time = 32.0881s    Reduced Chi^2 = 2.1367      Levenberg-Marquardt = 0.1
 Free parameter values:
        9.892         0.6216          41.05             20          18.98          0.769          34.11
-------------------------------------------------------------------------------
Iteration = 2
------------------
 Total time = 57.6663s    Reduced Chi^2 = 1.1212      Levenberg-Marquardt = 0.01
 Free parameter values:
       -0.1995         0.3908          35.33             20          25.61         0.8426          63.52
-------------------------------------------------------------------------------
Iteration = 3
------------------
 Total time = 83.0471s    Reduced Chi^2 = 0.99309      Levenberg-Marquardt = 0.001
 Free parameter values:
       -0.4497          0.479          35.34             20          23.77         0.9306          32.63
-------------------------------------------------------------------------------
Iteration = 4
------------------
 Total time = 108.2029s    Reduced Chi^2 = 0.91406      Levenberg-Marquardt = 0.0001
 Free parameter values:
       -0.7092         0.4732          35.11             20          22.08         0.9027          45.89
-------------------------------------------------------------------------------
Iteration = 5
------------------
 Total time = 133.0697s    Reduced Chi^2 = 0.91999      Levenberg-Marquardt = 1e-05
 Free parameter values:
       -0.1352         0.4653          35.16             20          19.99         0.8787          43.86
 Total time = 138.1739s    Reduced Chi^2 = 0.91865      Levenberg-Marquardt = 0.0001
 Free parameter values:
       -0.1352         0.4653          35.16             20          19.99         0.8787          43.86
 Total time = 143.169s    Reduced Chi^2 = 0.91912      Levenberg-Marquardt = 0.001
 Free parameter values:
       -0.1356         0.4653          35.16             20             20         0.8789          43.88
 Total time = 148.2226s    Reduced Chi^2 = 0.92066      Levenberg-Marquardt = 0.1
 Free parameter values:
```

```
      -0.1685          0.466          35.15             20           20.5        0.8877          44.61
 Total time = 153.167s    Reduced Chi^2 = 0.91459      Levenberg-Marquardt = 10
 Free parameter values:
      -0.5464         0.4712          35.12             20          21.75        0.9018          45.63
 Total time = 158.2901s    Reduced Chi^2 = 0.92236      Levenberg-Marquardt = 1000
 Free parameter values:
      -0.6922          0.473          35.11             20          22.05        0.9026          45.87
 *** No improvement in chi-squared over previous iteration ***
 Total time = 158.3003s    Reduced Chi^2 = 0.91406
 Free parameter values:
      -0.7092         0.4732          35.11             20          22.08        0.9027          45.89
-------------------------------------------------------------------------------
Fit converged:

Parameter values (free parameters with error estimates):
-----------------------------------------------------

Foreground parameter values:
----------------------------
  Function 1:
    1              1
    2              8
    3        -0.7092   +/-   2.088
    4         0.4732   +/-   0.06673
    5              0
    6          35.11   +/-   0.2445
    7             20   +/-   61.36
    8              0
    9              0
   10              0

  Function 2:
    1              1
    2              8
    3          22.08   +/-   8.796
    4         0.9027   +/-   0.1522
    5              0
    6          35.11   +/-   0.2445            bound to parameter 6 of foreground 1
    7             20   +/-   61.36              bound to parameter 7 of foreground 1
    8              0
    9              0
   10              0

  Function 3:
    1              1
    2              8
    3          45.89   +/-   7.521
    4          1.299   +/-   0.1617
    5              0
    6          35.11   +/-   0.2445            bound to parameter 6 of foreground 1
    7             20   +/-   61.36              bound to parameter 7 of foreground 1
    8              0
    9              0
   10              0

  Function 4:
    1              1
    2              8
    3          49.13   +/-   5.549
    4          1.235   +/-   0.1096
    5              0
    6          35.11   +/-   0.2445            bound to parameter 6 of foreground 1
    7             20   +/-   61.36              bound to parameter 7 of foreground 1
    8              0
    9              0
   10              0

  Function 5:
```

```
    1             1
    2             8
    3         63.29   +/-   4.803
    4         1.424   +/-   0.08322
    5             0
    6         35.11   +/-   0.2445            bound to parameter 6 of foreground 1
    7            20   +/-   61.36             bound to parameter 7 of foreground 1
    8             0
    9             0
   10             0


Background parameter values:
---------------------------
  Function 1:
    1         0.273   +/-   0.00478
    2             0

  Function 2:
    1         0.117   +/-   0.004461
    2             0

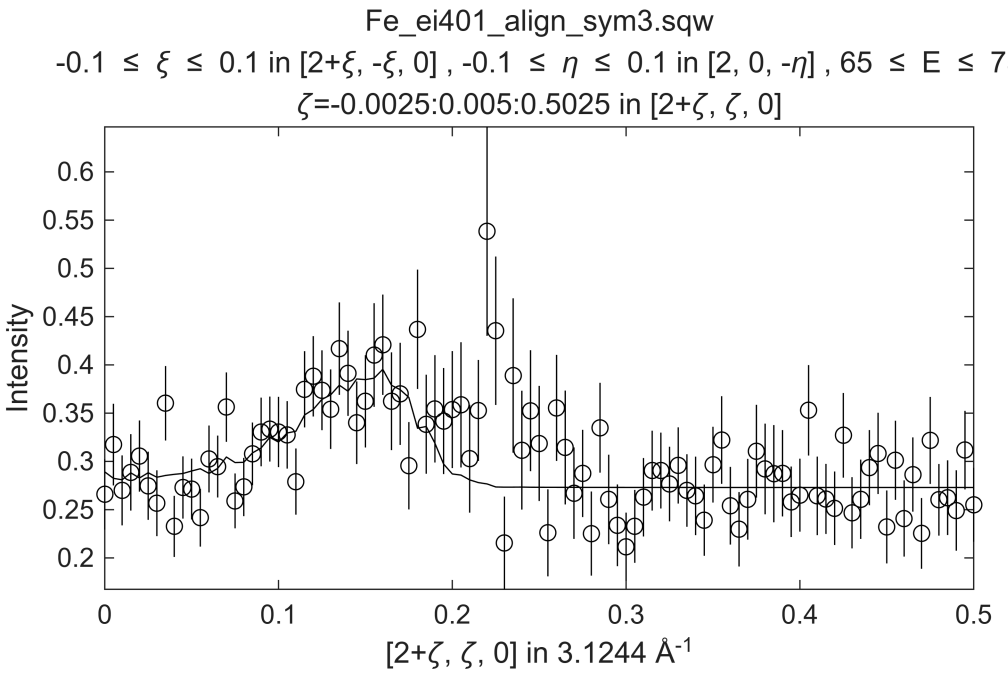
  Function 3:
    1       0.02657   +/-   0.003104
    2             0

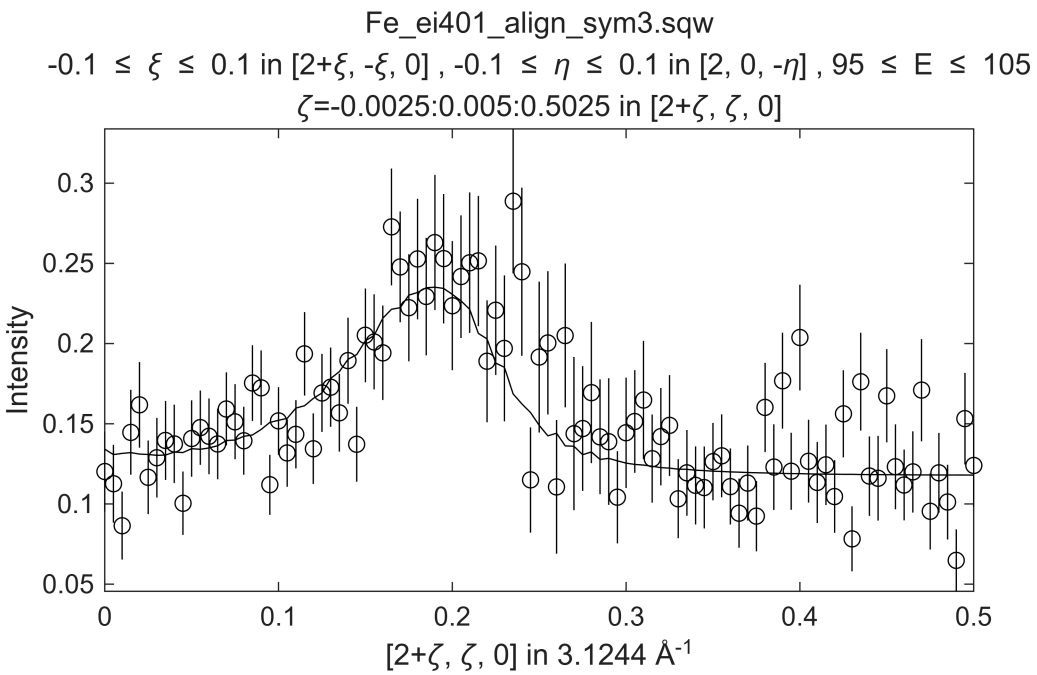
  Function 4:
    1       0.01903   +/-   0.002098
    2             0

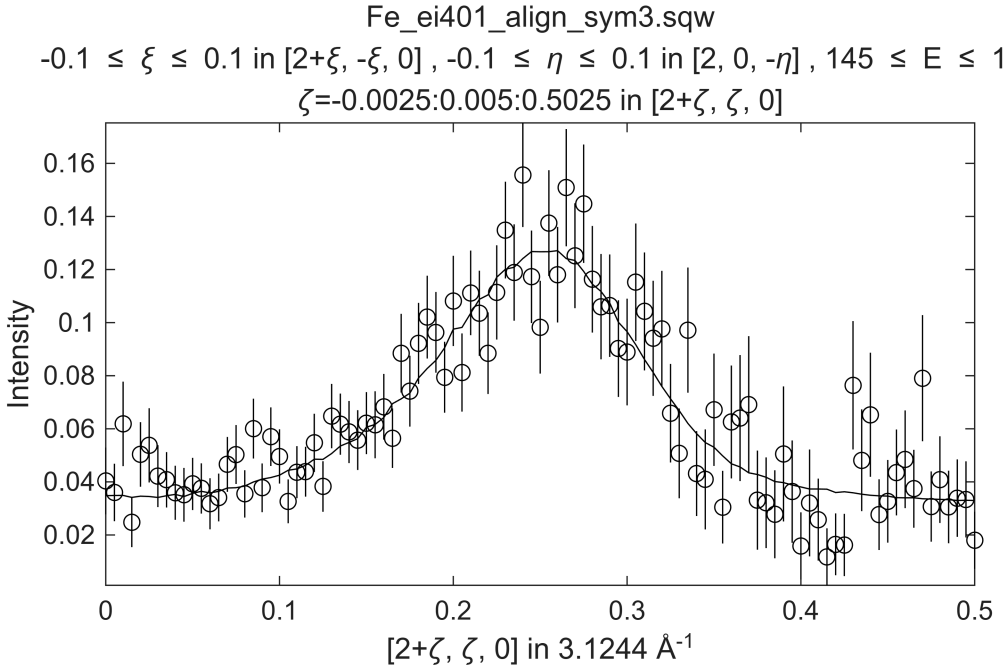
  Function 5:
    1       0.01132   +/-   0.001064
    2             0

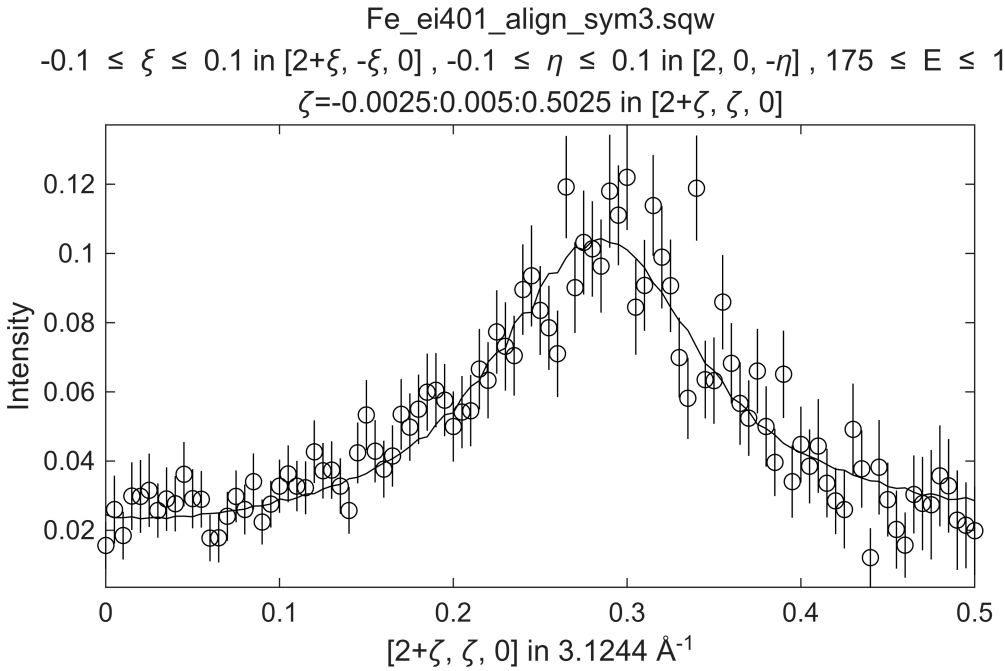

Covariance matrix for free parameters:
--------------------------------------
    1.0000   -0.3850   -0.0449    0.9813   -0.0196   -0.0015   -0.0299   -0.0167   -0.0155   -0.0081   -0.0076
   -0.3850    1.0000   -0.0585   -0.4133    0.0140    0.0035    0.0059   -0.0002   -0.0155   -0.0175   -0.0258
   -0.0449   -0.0585    1.0000   -0.0667   -0.0122   -0.0288    0.1573    0.1275    0.2919    0.2596    0.3491
    0.9813   -0.4133   -0.0667    1.0000   -0.1880   -0.1579   -0.0918   -0.0733   -0.0536   -0.0398   -0.0327
   -0.0196    0.0140   -0.0122   -0.1880    1.0000    0.8649   -0.0000   -0.0007   -0.0034   -0.0035   -0.0049
   -0.0015    0.0035   -0.0288   -0.1579    0.8649    1.0000   -0.0043   -0.0036   -0.0084   -0.0075   -0.0102
   -0.0299    0.0059    0.1573   -0.0918   -0.0000   -0.0043    1.0000    0.9157    0.0461    0.0405    0.0542
   -0.0167   -0.0002    0.1275   -0.0733   -0.0007   -0.0036    0.9157    1.0000    0.0373    0.0329    0.0442
   -0.0155   -0.0155    0.2919   -0.0536   -0.0034   -0.0084    0.0461    0.0373    1.0000    0.8706    0.1018
   -0.0081   -0.0175    0.2596   -0.0398   -0.0035   -0.0075    0.0405    0.0329    0.8706    1.0000    0.0907
   -0.0076   -0.0258    0.3491   -0.0327   -0.0049   -0.0102    0.0542    0.0442    0.1018    0.0907    1.0000
   -0.0106   -0.0231    0.3410   -0.0296   -0.0046   -0.0099    0.0532    0.0433    0.0995    0.0886    0.8418
    0.2194   -0.5702   -0.0027    0.2322   -0.0050   -0.0006   -0.0062   -0.0031   -0.0014    0.0002    0.0011
   -0.0002   -0.0005    0.0075    0.1207   -0.6823   -0.8170    0.0012    0.0010    0.0022    0.0019    0.0026
    0.0146    0.0007   -0.1177    0.0578    0.0007    0.0033   -0.8039   -0.9064   -0.0344   -0.0304   -0.0408
    0.0058    0.0135   -0.1962    0.0253    0.0026    0.0057   -0.0306   -0.0249   -0.7516   -0.8893   -0.0686
    0.0109    0.0168   -0.2724    0.0174    0.0034    0.0079   -0.0427   -0.0347   -0.0795   -0.0707   -0.6817
```

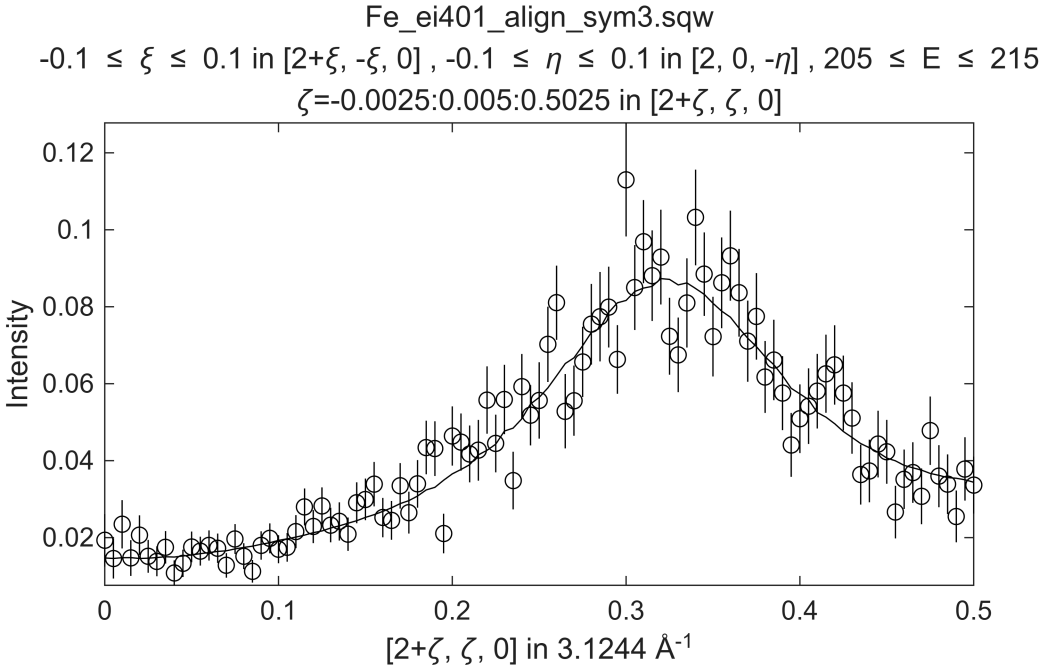
```
for i=1:numel(cuts)
    plot(cuts{i});
    pl(fit_obj{i}); keep_figure
end
```


Fe_ei401_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [2+ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, -η] , 65 ≤ E ≤ 75
ζ=-0.0025:0.005:0.5025 in [2+ζ, ζ, 0]
Intensity
0.6
0.55
0.5
0.45
0.4
0.35
0.3
0.25
0.2
0
0.1
0.2
0.3
0.4
0.5
[2+ζ, ζ, 0] in 3.1244 Å-1


Fe_ei401_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [2+ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, -η] , 95 ≤ E ≤ 105
ζ=-0.0025:0.005:0.5025 in [2+ζ, ζ, 0]
Intensity
0.3
0.25
0.2
0.15
0.1
0.05
0
0.1
0.2
0.3
0.4
0.5
[2+ζ, ζ, 0] in 3.1244 Å-1


Fe_ei401_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [2+ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, -η] , 145 ≤ E ≤ 155
ζ=-0.0025:0.005:0.5025 in [2+ζ, ζ, 0]
Intensity
0.16
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
0.1
0.2
0.3
0.4
0.5
[2+ζ, ζ, 0] in 3.1244 Å-1


Fe_ei401_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [2+ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, -η] , 175 ≤ E ≤ 185
ζ=-0.0025:0.005:0.5025 in [2+ζ, ζ, 0]
Intensity
0.12
0.1
0.08
0.06
0.04
0.02
0
0.1
0.2
0.3
0.4
0.5
[2+ζ, ζ, 0] in 3.1244 Å-1


Fe_ei401_align_sym3.sqw
-0.1 ≤ ξ ≤ 0.1 in [2+ξ, -ξ, 0] , -0.1 ≤ η ≤ 0.1 in [2, 0, -η] , 205 ≤ E ≤ 215
ζ=-0.0025:0.005:0.5025 in [2+ζ, ζ, 0]
Intensity
0.12
0.1
0.08
0.06
0.04
0.02
0
0.1
0.2
0.3
0.4
0.5
[2+ζ, ζ, 0] in 3.1244 Å-1